

From Reactive Creative Testing to Proactive Creative Intelligence

Core Insight

The challenge is not predicting CTR —
it is deciding *what to do next* when information is incomplete and constantly changing.

Creative Fatigue

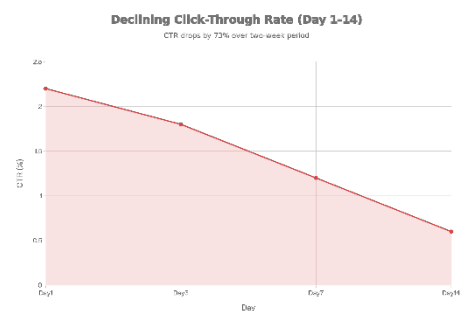
Problem:

CTR declines as users repeatedly see the same creative.

Observed Pattern (Supportive Data):

Solution:

- Monitor frequency exposure and decay slope
- Detect early fatigue signals
- Trigger proactive refresh before performance collapse
- Regenerate creatives using learned successful patterns



Outcome:

Sustained engagement and reduced wasted impressions.

Delayed Feedback Loops

Problem:

Reliable performance metrics arrive only after significant budget has been spent.

Our Approach:

- Leverage early proxy signals (engagement, watch time)
- Predict performance as a CTR range, not a point estimate
- Continuously update predictions as new data arrives

Example Output:

Expected CTR: 1.0% – 1.6% | Confidence: Medium

Outcome:

Faster, risk-aware creative decisions.

Attribution Noise

Attribution Noise → CTR depends on many things

What this means

CTR is affected by much more than the creative itself.

Real-life example (same ad)

Situation	CTR
Festival sale	2.5%
Normal week	1.1%
High bid	2.0%
Low bid	0.9%

The model rewards:

- High bids
- Good timing
- Easy audiences

Instead of rewarding good creative design.

Creative Intelligence Engine response:

- Separates creative impact from context
- Uses causal reasoning:

“If everything else stayed same, would this creative still win?”

Difference:

- Reactive → pick high CTR
- Proactive → pick causally strong creative

NON-STATIONARY DATA → Past ≠ Future

Meaning:

Patterns change over time.

Real Example:

- Bright colors worked last year
- This year → minimalistic designs perform better

Why CTR prediction fails:

Traditional ML assumes:

“Train on past → predict future”

But marketing trends shift.

Creative Intelligence Engine response:

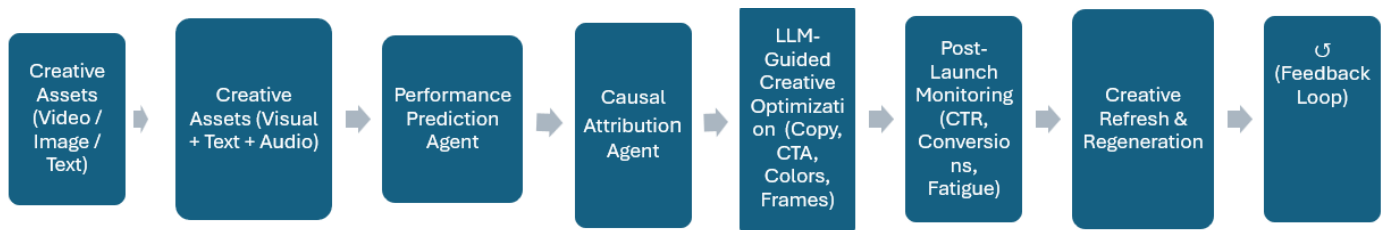
- Monitors drift
- Discounts old data
- Relearns patterns continuously

📌 Difference:

- Reactive → model becomes outdated
- Proactive → model evolves

How the Agentic System Works (End-to-End)

Agent-Oriented Architecture



Multimodal Understanding Agent

What it does

This agent understands *what the creative contains* and *how it communicates*.

For videos, images, and text, it extracts:

- Visual signals (faces, objects, colors)
- Text signals (sentiment, CTA clarity)
- Audio signals (tone, energy)
- Temporal signals (hook timing, pacing)

Output:

A structured representation of the creative (“Creative DNA”).

Performance Prediction Agent (Pre-Launch)

Problem Solved

Delayed feedback loops and early-stage uncertainty.

How it works

- Uses Creative DNA + historical patterns
- Predicts CTR as a range, not a single value

Example Output

Expected CTR: 1.0% – 1.6%

Risk Level: Medium

Why this helps

- Enables early decisions
- Reduces wasted spend
- Supports risk-aware testing

Causal Attribution Agent

Problem Solved

Attribution noise and misleading correlations.

How it works

- Separates creative impact from context (bid, targeting, season)
- Learns which creative elements *actually cause* performance changes

Example Insight

- Early visual hook → positive impact
- Dense text overlays → negative impact

This ensures teams learn the right lessons.

Creative Optimization Agent (LLM-Guided)

Problem Solved

Slow and manual creative iteration.

How it works

The agent converts insights into clear, actionable suggestions:

- Rewrite copy
- Change CTA placement
- Adjust color palette
- Reorder video frames

These are recommendations, not automatic changes.

Human Review & Control Agent

Why humans remain essential

- **Brand tone**
- **Contextual judgment**
- **Compliance and trust**

Humans:

- **Approve**
- **Edit**
- **Reject AI suggestions**

Their decisions become learning signals for the system.

Monitoring & Learning Agent

Problems Solved

Creative fatigue and non-stationary behavior.

How it works

- **Tracks performance trends and exposure frequency**
- **Detects early fatigue signals**
- **Triggers refresh before performance collapses**
- **Learns continuously from outcomes**